

T3 matched EFT Lagrangian terms: EFT_1_after_F

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. This stage integrates out F. Remaining active T3 fields: S1, S2.

Field combinations 1 of 6

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$B_{\mu\nu}$	D	F	H
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$	$\begin{aligned} & +\frac{2}{3}\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)\frac{\omega^2}{g_{2,\tau_1,\tau_1}g_{2,\tau_2,\tau_2}S_{2,j_2}g_{2,\tau_1,\tau_2}g_{2,\tau_2,\tau_1}}\left(S_{2,j_1}^\dagger\right)^2 \\ & -\frac{2C\text{N e w S c a l a r }22\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)}{\sqrt{3}}\frac{g_{2,\tau_1,\tau_1}^2g_{2,j_1}g_{2,\tau_1,\tau_1}S_{2,j_2}g_{2,\tau_1,\tau_2}g_{2,\tau_2,\tau_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{\sqrt{3}} \\ & -\frac{2\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)}{\sqrt{3}}\frac{g_{2,\tau_1,\tau_1}^2g_{2,j_2}S_{2,j_4}g_{2,\tau_1,\tau_1}S_{2,j_2}^\dagger S_{2,j_2}^\dagger T_{S_2,j_2,j_1,j_4,j_2}^\dagger\lambda_{S_2}^1}{\sqrt{3}} \\ & -\frac{2\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)}{\sqrt{3}}\frac{g_{2,\tau_1,\tau_1}^2g_{2,j_2}S_{2,j_4}g_{2,\tau_1,\tau_1}S_{2,j_2}^\dagger S_{2,j_2}^\dagger T_{S_2,j_2,j_1,j_4,j_2}^2\lambda_{S_2}^2}{\sqrt{3}} \end{aligned}$	$\begin{aligned} & +\frac{2\mathcal{M}_{\tau_1,\tau_1}^2g_{1,\tau_2,\tau_2}^2L_{\tau_1,\tau_2}^2}{\sqrt{6}M_F}C\cdot P_{\tau_1}L_{\tau_2,j_1}H_{\tau_1}H_{\tau_2}\lambda_{\tau_1}\text{Bar}\left(\exp\left(S\cdot U\cdot L\right)\right)_{\tau_1,\tau_2}\text{Bar}\left(\exp\left(S\cdot U\cdot L\right)\right)_{\tau_1,\tau_2} \\ & +\frac{2\mathcal{M}_{\tau_1,\tau_1}^2g_{1,\tau_2,\tau_2}^2L_{\tau_1,\tau_2}^2}{\sqrt{6}M_F}C\cdot P_{\tau_1}L_{\tau_2,j_1}H_{\tau_1}H_{\tau_2}\lambda_{\tau_1}\text{Bar}\left(\exp\left(S\cdot U\cdot L\right)\right)_{\tau_1,\tau_2}\text{Bar}\left(\exp\left(S\cdot U\cdot L\right)\right)_{\tau_1,\tau_2} \\ & +\frac{2\mathcal{M}_{\tau_1,\tau_1}^2g_{1,\tau_2,\tau_2}^2C\cdot P_{\tau_1}L_{\tau_2,\tau_2}g_{1,\tau_1,\tau_1}g_{2,\tau_2,\tau_2}\exp\left(S\cdot U\cdot L\right)_{\tau_1,\tau_2}\exp\left(S\cdot U\cdot L\right)_{\tau_2,\tau_2}H_{\tau_1}^\dagger H_{\tau_2}^\dagger}{\sqrt{6}M_F} \\ & +\frac{2\mathcal{M}_{\tau_1,\tau_1}^2g_{1,\tau_2,\tau_2}^2C\cdot P_{\tau_1}L_{\tau_2,\tau_2}g_{1,\tau_1,\tau_1}g_{2,\tau_2,\tau_2}\exp\left(S\cdot U\cdot L\right)_{\tau_1,\tau_2}\exp\left(S\cdot U\cdot L\right)_{\tau_2,\tau_2}H_{\tau_1}^\dagger H_{\tau_2}^\dagger}{\sqrt{6}M_F} \\ & -\frac{2\mathcal{M}_{\tau_1,\tau_1}^2g_{1,\tau_2,\tau_2}^2Y_{\tau_1,\tau_2,j_1}H_{\tau_1}S_{2,j_2}g_{2,\tau_2,\tau_2}Y_{\tau_1,\tau_2}H_{\tau_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}} \\ & -\frac{2\mathcal{M}_{\tau_1,\tau_1}^2g_{1,\tau_2,\tau_2}^2Y_{\tau_1,\tau_2,j_1}H_{\tau_1}S_{2,j_2}g_{2,\tau_2,\tau_2}Y_{\tau_1,\tau_2}H_{\tau_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}} \\ & -\frac{2i\sqrt{\frac{2}{3}}\mathcal{M}_{\tau_1,\tau_1}^2g_{1,\tau_2,\tau_2}^2H_{\tau_2}S_{2,j_2}g_{2,\tau_2,\tau_2}Y_{\tau_1,\tau_1}H_{\tau_1}^\dagger S_{2,j_1}^\dagger T_{\tau_1,\tau_1,\tau_2,\tau_2,j_1}}{\sqrt{3}} \\ & -2i\sqrt{\frac{2}{3}}\mathcal{M}_{\tau_1,\tau_1}^2g_{1,\tau_2,\tau_2}^2H_{\tau_2}S_{2,j_2}g_{2,\tau_2,\tau_2}Y_{\tau_1,\tau_1}H_{\tau_1}^\dagger S_{2,j_1}^\dagger T_{\tau_1,\tau_1,\tau_2,\tau_2,j_1}} \\ & +H_{\tau_2}S_{2,j_2}H_{\tau_1}^\dagger S_{2,j_1}^\dagger T_{\tau_1,\tau_1,\tau_2,\tau_2,j_1}^2\lambda_{\tau_1}^2 \\ & -\frac{2\mathcal{M}_{\tau_1,\tau_1}^2g_{1,\tau_2,\tau_2}^2H_{\tau_2}S_{2,j_2}g_{2,\tau_2,\tau_2}H_{\tau_1}^\dagger S_{2,j_1}^\dagger T_{\tau_1,\tau_1,\tau_2,\tau_2,j_1}}{\sqrt{3}}^2\lambda_{\tau_1}^2 \\ & -\frac{4\mathcal{M}_{\tau_1,\tau_1}^2g_{1,\tau_2,\tau_2}^2H_{\tau_2}S_{2,j_2}g_{2,\tau_2,\tau_2}H_{\tau_1}^\dagger S_{2,j_1}^\dagger T_{\tau_1,\tau_1,\tau_2,\tau_2,j_1}}{\sqrt{3}}^2\lambda_{\tau_1}^2 \end{aligned}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$\begin{aligned} & +\frac{2}{3}\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)\frac{\omega^2}{g_{1,\tau_1,\tau_1}g_{1,\tau_2,\tau_2}S_{1,j_2}g_{1,\tau_1,\tau_2}g_{1,\tau_2,\tau_1}}\left(S_{1,j_1}^\dagger\right)^2 \\ & -\frac{2C\text{N e w S c a l a r }12\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)}{\sqrt{3}}\frac{g_{1,\tau_1,\tau_1}^2g_{1,j_1}S_{1,j_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}} \\ & -\frac{4}{3}\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)\frac{g_{1,\tau_1,\tau_1}^2g_{1,j_2}S_{1,j_4}g_{1,\tau_1,\tau_1}S_{1,j_2}^\dagger S_{1,j_2}^\dagger T_{S_1,j_2,j_1,j_4,j_2}^\dagger\lambda_{S_1}^1}{\sqrt{3}} \\ & -\frac{2\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)}{\sqrt{3}}\frac{g_{1,\tau_1,\tau_1}^2g_{1,j_2}S_{1,j_4}g_{1,\tau_1,\tau_1}S_{1,j_2}^\dagger S_{1,j_2}^\dagger T_{S_1,j_2,j_1,j_4,j_2}^2\lambda_{S_1}^2}{\sqrt{3}} \end{aligned}$	$\begin{aligned} & +\frac{2\mathcal{M}_{\tau_1,\tau_1}^2g_{1,\tau_2,\tau_2}^2L_{\tau_1,\tau_2}^2}{\sqrt{6}M_F}C\cdot P_{\tau_1}L_{\tau_2,j_1}H_{\tau_1}H_{\tau_2}\lambda_{\tau_1}\text{Bar}\left(\exp\left(S\cdot U\cdot L\right)\right)_{\tau_1,\tau_2}\text{Bar}\left(\exp\left(S\cdot U\cdot L\right)\right)_{\tau_1,\tau_2} \\ & +\frac{2\mathcal{M}_{\tau_1,\tau_1}^2g_{1,\tau_2,\tau_2}^2L_{\tau_1,\tau_2}^2}{\sqrt{6}M_F}C\cdot P_{\tau_1}L_{\tau_2,j_1}H_{\tau_1}H_{\tau_2}\lambda_{\tau_1}\text{Bar}\left(\exp\left(S\cdot U\cdot L\right)\right)_{\tau_1,\tau_2}\text{Bar}\left(\exp\left(S\cdot U\cdot L\right)\right)_{\tau_1,\tau_2} \\ & +\frac{2\mathcal{M}_{\tau_1,\tau_1}^2g_{1,\tau_2,\tau_2}^2C\cdot P_{\tau_1}L_{\tau_2,\tau_2}g_{1,\tau_1,\tau_1}g_{2,\tau_2,\tau_2}\exp\left(S\cdot U\cdot L\right)_{\tau_1,\tau_2}\exp\left(S\cdot U\cdot L\right)_{\tau_2,\tau_2}H_{\tau_1}^\dagger H_{\tau_2}^\dagger}{\sqrt{6}M_F} \\ & +\frac{2\mathcal{M}_{\tau_1,\tau_1}^2g_{1,\tau_2,\tau_2}^2C\cdot P_{\tau_1}L_{\tau_2,\tau_2}g_{1,\tau_1,\tau_1}g_{2,\tau_2,\tau_2}\exp\left(S\cdot U\cdot L\right)_{\tau_1,\tau_2}\exp\left(S\cdot U\cdot L\right)_{\tau_2,\tau_2}H_{\tau_1}^\dagger H_{\tau_2}^\dagger}{\sqrt{6}M_F} \\ & -\frac{2\mathcal{M}_{\tau_1,\tau_1}^2g_{1,\tau_2,\tau_2}^2Y_{\tau_1,\tau_2,j_1}H_{\tau_1}S_{1,j_2}g_{2,\tau_2,\tau_2}Y_{\tau_1,\tau_2}H_{\tau_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}} \\ & -\frac{2\mathcal{M}_{\tau_1,\tau_1}^2g_{1,\tau_2,\tau_2}^2Y_{\tau_1,\tau_2,j_1}H_{\tau_1}S_{1,j_2}g_{2,\tau_2,\tau_2}Y_{\tau_1,\tau_2}H_{\tau_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}} \\ & -\frac{2i\sqrt{\frac{2}{3}}\mathcal{M}_{\tau_1,\tau_1}^2g_{1,\tau_2,\tau_2}^2H_{\tau_2}S_{1,j_2}g_{2,\tau_2,\tau_2}Y_{\tau_1,\tau_1}H_{\tau_1}^\dagger S_{1,j_1}^\dagger T_{\tau_1,\tau_1,\tau_2,\tau_2,j_1}}{\sqrt{3}} \\ & -2i\sqrt{\frac{2}{3}}\mathcal{M}_{\tau_1,\tau_1}^2g_{1,\tau_2,\tau_2}^2H_{\tau_2}S_{1,j_2}g_{2,\tau_2,\tau_2}Y_{\tau_1,\tau_1}H_{\tau_1}^\dagger S_{1,j_1}^\dagger T_{\tau_1,\tau_1,\tau_2,\tau_2,j_1}} \\ & +H_{\tau_2}S_{1,j_2}H_{\tau_1}^\dagger S_{1,j_1}^\dagger T_{\tau_1,\tau_1,\tau_2,\tau_2,j_1}^2\lambda_{\tau_1}^2 \\ & -\frac{2\mathcal{M}_{\tau_1,\tau_1}^2g_{1,\tau_2,\tau_2}^2H_{\tau_2}S_{1,j_2}g_{2,\tau_2,\tau_2}H_{\tau_1}^\dagger S_{1,j_1}^\dagger T_{\tau_1,\tau_1,\tau_2,\tau_2,j_1}}{\sqrt{3}}^2\lambda_{\tau_1}^2 \\ & -\frac{4\mathcal{M}_{\tau_1,\tau_1}^2g_{1,\tau_2,\tau_2}^2H_{\tau_2}S_{1,j_2}g_{2,\tau_2,\tau_2}H_{\tau_1}^\dagger S_{1,j_1}^\dagger T_{\tau_1,\tau_1,\tau_2,\tau_2,j_1}}{\sqrt{3}}^2\lambda_{\tau_1}^2 \end{aligned}$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$B_{\mu\nu}$	D	F	H
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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	$W_{\mu\nu}$	constant
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Field combinations 3 of 6

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F B_{\mu\nu}$	$F H$	$F W_{\mu\nu}$	$H e$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)B_{\mu_1\mu_2}^2$	$+\frac{2\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)}{\sqrt{M_F}}\frac{\overline{\text{tr}}_{\tau_1,\tau_2}\left(\text{tr}_{\tau_2,\tau_1}\right)^{\dagger}L_{a_1,i_2}{}^TC P_L L_{a_2,i_1}H_{a_1}H_{a_2}Bw\otimes\text{pr}(\text{SU}(L))_{\tau_1,a_2}Bw\otimes\text{pr}(\text{SU}(L))_{\tau_2,a_1}}{\sqrt{M_F}}$ $+\frac{2\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)}{\sqrt{M_F}}\frac{\overline{\text{tr}}_{\tau_1}\delta_{a_2,i_1}{}^TC P_R\overline{\text{tr}}_{a_2,i_2}{}^T\overline{\text{tr}}_{\tau_1,\tau_2}\left(\text{tr}_{\tau_2,\tau_1}\right)\text{pr}(\text{SU}(L))_{\tau_1,a_2}\text{pr}(\text{SU}(L))_{\tau_2,a_1}H_{a_1}{}^{\dagger}}{\sqrt{M_F}}$ $+2i\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\frac{\overline{\text{tr}}_{\tau_1,\tau_2}\left(\text{tr}_{\tau_2,\tau_1}H_{a_1}S_{a_1}\overline{\text{tr}}_{a_2,i_2}{}^TC_{a_1,i_1}H_{a_1}{}^{\dagger}S_{a_1}{}^{\dagger}\right)}{\sqrt{3}}$ $-\frac{2\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)}{\sqrt{3}}\frac{\overline{\text{tr}}_{\tau_1,\tau_2}H_{a_2}S_{a_2}\overline{\text{tr}}_{\tau_1,\tau_2}H_{a_1}{}^{\dagger}S_{a_1}{}^{\dagger}\overline{\text{tr}}_{a_1,-a_2,i_2,i_1}{}^{\dagger}\text{tr}_{a_2,-a_1,i_2,i_1}{}^{\dagger}}{\sqrt{3}}$	$-\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_2}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_2}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_2}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{h\overline{y}_{1,i_1,r_1}\overline{Y}_{e,i_3,i_2}\overline{e}_{i_2}}{4\epsilon}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}{}^{\dagger}$ $+\frac{3}{8}h\overline{y}_{1,i_1,r_1}\overline{Y}_{e,i_3,i_2}\overline{e}_{i_2}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}{}^{\dagger}$ $+\frac{\sqrt{3}h\overline{y}_{2,i_1,r_1}\overline{Y}_{e,i_3,i_2}\overline{e}_{i_2}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^{\dagger}$ $+\frac{3}{8}\sqrt{3}h\overline{y}_{2,i_1,r_1}\overline{Y}_{e,i_3,i_2}\overline{e}_{i_2}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^{\dagger}$ $+\frac{\sqrt{3}h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_2}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}\sqrt{3}h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_2}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_2}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_2}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1}\overline{Y}_{e,i_3,i_2}\overline{e}_{i_2}}{4\epsilon}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}{}^{\dagger}$ $+\frac{h\overline{y}_{2,i_1,r_1}\overline{Y}_{e,i_3,i_2}\overline{e}_{i_2}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^{\dagger}$ $+\frac{3}{8}\sqrt{3}h\overline{y}_{1,i_1,r_1}\overline{Y}_{e,i_3,i_2}\overline{e}_{i_2}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}{}^{\dagger}$ $+\frac{h\overline{y}_{2,i_1,r_1}\overline{Y}_{e,i_3,i_2}\overline{e}_{i_2}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^{\dagger}$ $+\frac{3}{8}h\overline{y}_{1,i_1,r_1}\overline{Y}_{e,i_3,i_2}\overline{e}_{i_2}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^{\dagger}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)B_{\mu_1\mu_2}^2$	$+\frac{2\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)}{\sqrt{M_F}}\frac{\overline{\text{tr}}_{\tau_1,\tau_2}\left(\text{tr}_{\tau_2,\tau_1}\right)^{\dagger}L_{a_1,i_2}{}^TC P_L L_{a_2,i_1}H_{a_1}H_{a_2}Bw\otimes\text{pr}(\text{SU}(L))_{\tau_1,a_2}Bw\otimes\text{pr}(\text{SU}(L))_{\tau_2,a_1}}{\sqrt{M_F}}$ $+\frac{2\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)}{\sqrt{M_F}}\frac{\overline{\text{tr}}_{\tau_1}\delta_{a_2,i_1}{}^TC P_R\overline{\text{tr}}_{a_2,i_2}{}^T\overline{\text{tr}}_{\tau_1,\tau_2}\left(\text{tr}_{\tau_2,\tau_1}\right)\text{pr}(\text{SU}(L))_{\tau_1,a_2}\text{pr}(\text{SU}(L))_{\tau_2,a_1}H_{a_1}{}^{\dagger}}{\sqrt{M_F}}$ $-2i\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\frac{\overline{\text{tr}}_{\tau_1,\tau_2}\left(\text{tr}_{\tau_2,\tau_1}H_{a_1}S_{a_1}\overline{\text{tr}}_{a_2,i_2}{}^TC_{a_1,i_1}H_{a_1}{}^{\dagger}S_{a_1}{}^{\dagger}\right)}{\sqrt{3}}$ $-\frac{2\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)}{\sqrt{3}}\frac{\overline{\text{tr}}_{\tau_1,\tau_2}H_{a_2}S_{a_2}\overline{\text{tr}}_{\tau_1,\tau_2}H_{a_1}{}^{\dagger}S_{a_1}{}^{\dagger}\overline{\text{tr}}_{a_1,-a_2,i_2,i_1}{}^{\dagger}\text{tr}_{a_2,-a_1,i_2,i_1}{}^{\dagger}}{\sqrt{3}}$	$-\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{\sqrt{3}h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_2}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}\sqrt{3}h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_2}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_2}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_2}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1}\overline{Y}_{e,i_3,i_2}\overline{e}_{i_2}}{4\epsilon}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}{}^{\dagger}$ $+\frac{h\overline{y}_{2,i_1,r_1}\overline{Y}_{e,i_3,i_2}\overline{e}_{i_2}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^{\dagger}$ $+\frac{3}{8}\sqrt{3}h\overline{y}_{1,i_1,r_1}\overline{Y}_{e,i_3,i_2}\overline{e}_{i_2}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}{}^{\dagger}$ $+\frac{h\overline{y}_{2,i_1,r_1}\overline{Y}_{e,i_3,i_2}\overline{e}_{i_2}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^{\dagger}$ $+\frac{3}{8}h\overline{y}_{1,i_1,r_1}\overline{Y}_{e,i_3,i_2}\overline{e}_{i_2}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^{\dagger}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)B_{\mu_1\mu_2}^2$	$+2i\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\frac{\overline{\text{tr}}_{\tau_1,\tau_2}\left(\text{tr}_{\tau_2,\tau_1}\right)^{\dagger}L_{a_1,i_2}{}^TC P_L L_{a_2,i_1}H_{a_1}H_{a_2}Bw\otimes\text{pr}(\text{SU}(L))_{\tau_1,a_2}Bw\otimes\text{pr}(\text{SU}(L))_{\tau_2,a_1}}{\sqrt{M_F}}$ $-2i\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\frac{\overline{\text{tr}}_{\tau_1}\delta_{a_2,i_1}{}^TC P_R\overline{\text{tr}}_{a_2,i_2}{}^T\overline{\text{tr}}_{\tau_1,\tau_2}\left(\text{tr}_{\tau_2,\tau_1}\right)\text{pr}(\text{SU}(L))_{\tau_1,a_2}\text{pr}(\text{SU}(L))_{\tau_2,a_1}H_{a_1}{}^{\dagger}}{\sqrt{M_F}}$ $-\frac{2\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)}{\sqrt{3}}\frac{\overline{\text{tr}}_{\tau_1,\tau_2}\left(\text{tr}_{\tau_2,\tau_1}H_{a_1}S_{a_1}\overline{\text{tr}}_{a_2,i_2}{}^TC_{a_1,i_1}H_{a_1}{}^{\dagger}S_{a_1}{}^{\dagger}\right)}{\sqrt{3}}$ $-\frac{2\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)}{\sqrt{3}}\frac{\overline{\text{tr}}_{\tau_1,\tau_2}H_{a_2}S_{a_2}\overline{\text{tr}}_{\tau_1,\tau_2}H_{a_1}{}^{\dagger}S_{a_1}{}^{\dagger}\overline{\text{tr}}_{a_1,-a_2,i_2,i_1}{}^{\dagger}\text{tr}_{a_2,-a_1,i_2,i_1}{}^{\dagger}}{\sqrt{3}}$ $-\frac{2\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)}{\sqrt{3}}\frac{\overline{\text{tr}}_{\tau_1,\tau_2}\left(\text{tr}_{\tau_2,\tau_1}H_{a_1}S_{a_1}\overline{\text{tr}}_{a_2,i_2}{}^TC_{a_1,i_1}H_{a_1}{}^{\dagger}S_{a_1}{}^{\dagger}\right)}{\sqrt{3}}$ $-\frac{2\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)}{\sqrt{3}}\frac{\overline{\text{tr}}_{\tau_1,\tau_2}H_{a_2}S_{a_2}\overline{\text{tr}}_{\tau_1,\tau_2}H_{a_1}{}^{\dagger}S_{a_1}{}^{\dagger}\overline{\text{tr}}_{a_1,-a_2,i_2,i_1}{}^{\dagger}\text{tr}_{a_2,-a_1,i_2,i_1}{}^{\dagger}}{\sqrt{3}}$ $-\frac{2\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)}{\sqrt{3}}\frac{\overline{\text{tr}}_{\tau_1,\tau_2}\left(\text{tr}_{\tau_2,\tau_1}H_{a_1}S_{a_1}\overline{\text{tr}}_{a_2,i_2}{}^TC_{a_1,i_1}H_{a_1}{}^{\dagger}S_{a_1}{}^{\dagger}\right)}{\sqrt{3}}$ $-\frac{2\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)}{\sqrt{3}}\frac{\overline{\text{tr}}_{\tau_1,\tau_2}H_{a_2}S_{a_2}\overline{\text{tr}}_{\tau_1,\tau_2}H_{a_1}{}^{\dagger}S_{a_1}{}^{\dagger}\overline{\text{tr}}_{a_1,-a_2,i_2,i_1}{}^{\dagger}\text{tr}_{a_2,-a_1,i_2,i_1}{}^{\dagger}}{\sqrt{3}}$ $-\frac{2\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)}{\sqrt{3}}\frac{\overline{\text{tr}}_{\tau_1,\tau_2}\left(\text{tr}_{\tau_2,\tau_1}H_{a_1}S_{a_1}\overline{\text{tr}}_{a_2,i_2}{}^TC_{a_1,i_1}H_{a_1}{}^{\dagger}S_{a_1}{}^{\dagger}\right)}{\sqrt{3}}$ $-\frac{2\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)}{\sqrt{3}}\frac{\overline{\text{tr}}_{\tau_1,\tau_2}H_{a_2}S_{a_2}\overline{\text{tr}}_{\tau_1,\tau_2}H_{a_1}{}^{\dagger}S_{a_1}{}^{\dagger}\overline{\text{tr}}_{a_1,-a_2,i_2,i_1}{}^{\dagger}\text{tr}_{a_2,-a_1,i_2,i_1}{}^{\dagger}}{\sqrt{3}}$	$-\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{\sqrt{3}h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_2}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}\sqrt{3}h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_2}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_2}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}\sqrt{3}h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_2}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{h\overline{y}_{1,i_1,r_1}\overline{Y}_{e,i_3,i_2}\overline{e}_{i_2}}{4\epsilon}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}{}^{\dagger}$ $+\frac{3}{8}h\overline{y}_{1,i_1,r_1}\overline{Y}_{e,i_3,i_2}\overline{e}_{i_2}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}{}^{\dagger}$ $+\frac{\sqrt{3}h\overline{y}_{2,i_1,r_1}\overline{Y}_{e,i_3,i_2}\overline{e}_{i_2}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^{\dagger}$ $+\frac{3}{8}\sqrt{3}h\overline{y}_{2,i_1,r_1}\overline{Y}_{e,i_3,i_2}\overline{e}_{i_2}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^{\dagger}$ $+\frac{h\overline{y}_{2,i_1,r_1}\overline{Y}_{e,i_3,i_2}\overline{e}_{i_2}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^{\dagger}$ $+\frac{3}{8}h\overline{y}_{2,i_1,r_1}\overline{Y}_{e,i_3,i_2}\overline{e}_{i_2}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^{\dagger}$

Field combinations 4 of 6

[illegible]

Field combinations 5 of 6

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2 F$	$S_2 H$	$F H e$	$S_1 F H$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0			$+\frac{1}{4}h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{1,i3,r1}L_{a_1,i2}}PR\,e_{i1}H_{a_1}y_{1,i2,r1}Y_{e,i3,i1}$ $+\frac{1}{4}h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{2,i3,r1}L_{a_1,i2}}PR\,e_{i1}H_{a_1}y_{2,i2,r1}Y_{e,i3,i1}$ $+\frac{1}{4}h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{1,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}}PL\,L_{a_1,i1}y_{1,i3,r1}H_{a_1}^\dagger$ $+\frac{1}{4}h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{2,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}}PL\,L_{a_1,i1}y_{2,i3,r1}H_{a_1}^\dagger$	
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0			$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{1,i3,r1}L_{a_1,i2}}PR\,e_{i1}H_{a_1}y_{1,i2,r1}Y_{e,i3,i1}$ $+\frac{1}{4}\sqrt{3}h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{2,i3,r1}L_{a_1,i2}}PR\,e_{i1}H_{a_1}y_{2,i2,r1}Y_{e,i3,i1}$ $+\frac{1}{4}\sqrt{3}h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{1,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}}PL\,L_{a_1,i1}y_{1,i3,r1}H_{a_1}^\dagger$ $+\frac{1}{4}\sqrt{3}h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{2,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}}PL\,L_{a_1,i1}y_{2,i3,r1}H_{a_1}^\dagger$	
T3-A	0	1	0	3	1	2	$\frac{1}{2}$			$+\frac{1}{4}h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{1,i3,r1}L_{a_1,i2}}PR\,e_{i1}H_{a_1}y_{1,i2,r1}Y_{e,i3,i1}$ $+\frac{1}{4}\sqrt{3}h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{2,i3,r1}L_{a_1,i2}}PR\,e_{i1}H_{a_1}y_{2,i2,r1}Y_{e,i3,i1}$ $+\frac{1}{4}h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{1,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}}PL\,L_{a_1,i1}y_{1,i3,r1}H_{a_1}^\dagger$ $+\frac{1}{4}\sqrt{3}h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{2,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}}PL\,L_{a_1,i1}y_{2,i3,r1}H_{a_1}^\dagger$	$-2h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{1,i1,r1}Y_{e,i2,i3}H_{a_1}S_1y_{1,i2,r1}Y_{e,i1,i3}H_{a_1}^\dagger S_1^\dagger}$ $-h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{1,i1,r1}\overline{\lambda_{T3}}S_{2,j1}y_{1,i1,r1}\text{Bar}(\text{T3MixCG})_{a_1,a2,j1}H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger}$ $-h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{2,i1,r1}\overline{\lambda_{T3}}S_{2,j1}y_{2,i1,r1}\text{Bar}(\text{T3MixCG})_{a_1,a2,j1}H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger}$ $-h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{1,i1,r1}H_{a_1}H_{a_2}S_1y_{1,i1,r1}\overline{\pi_{T3,a1,a2,j1}}\overline{\lambda_{T3}}S_{2,j1}^\dagger}$ $-h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{2,i1,r1}H_{a_1}H_{a_2}S_1y_{2,i1,r1}\overline{\pi_{T3,a1,a2,j1}}\overline{\lambda_{T3}}S_{2,j1}^\dagger}$ $-2h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{1,i1,r1}H_{a_2}S_1y_{1,i1,r1}H_{a_1}^\dagger S_1^\dagger\overline{\lambda_{H1,a1,a2}^{-1}}\overline{\lambda_{H1}^\dagger}}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$			$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{1,i3,r1}L_{a_1,i2}}PR\,e_{i1}H_{a_1}y_{1,i2,r1}Y_{e,i3,i1}$ $+\frac{1}{4}h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{2,i3,r1}L_{a_1,i2}}PR\,e_{i1}H_{a_1}y_{2,i2,r1}Y_{e,i3,i1}$ $+\frac{1}{4}\sqrt{3}h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{1,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}}PL\,L_{a_1,i1}y_{1,i3,r1}H_{a_1}^\dagger$ $+\frac{1}{4}h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{2,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}}PL\,L_{a_1,i1}y_{2,i3,r1}H_{a_1}^\dagger$	$-2h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{1,i1,r1}Y_{e,i2,i3}H_{a_1}S_1y_{1,i2,r1}Y_{e,i1,i3}H_{a_1}^\dagger S_1^\dagger}$ $-h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{1,i1,r1}\overline{\lambda_{T3}}S_{2,j1}y_{1,i1,r1}\text{Bar}(\text{T3MixCG})_{a_1,a2,j1}H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger}$ $-h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{2,i1,r1}\overline{\lambda_{T3}}S_{2,j1}y_{2,i1,r1}\text{Bar}(\text{T3MixCG})_{a_1,a2,j1}H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger}$ $-h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{1,i1,r1}H_{a_1}H_{a_2}S_1y_{1,i1,r1}\overline{\pi_{T3,a1,a2,j1}}\overline{\lambda_{T3}}S_{2,j1}^\dagger}$ $-h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{2,i1,r1}H_{a_1}H_{a_2}S_1y_{2,i1,r1}\overline{\pi_{T3,a1,a2,j1}}\overline{\lambda_{T3}}S_{2,j1}^\dagger}$ $-2h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{1,i1,r1}H_{a_2}S_1y_{1,i1,r1}H_{a_1}^\dagger S_1^\dagger\overline{\lambda_{H1,a1,a2}^{-1}}\overline{\lambda_{H1}^\dagger}}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$			$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{1,i3,r1}L_{a_1,i2}}PR\,e_{i1}H_{a_1}y_{1,i2,r1}Y_{e,i3,i1}$ $+\frac{1}{4}\sqrt{3}h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{2,i3,r1}L_{a_1,i2}}PR\,e_{i1}H_{a_1}y_{2,i2,r1}Y_{e,i3,i1}$ $+\frac{1}{4}\sqrt{3}h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{1,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}}PL\,L_{a_1,i1}y_{1,i3,r1}H_{a_1}^\dagger$ $+\frac{1}{4}\sqrt{3}h\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\overline{y_{2,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}}PL\,L_{a_1,i1}y_{2,i3,r1}H_{a_1}^\dagger$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_2	F	H
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0			
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0			
T3-A	0	1	0	3	1	2	$\frac{1}{2}$			
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{h \log \left(\frac{\mu b m_2^2}{M_F^2} \right) \overline{y_{1,i_1,\tau_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{1,i_1,\tau_1} \overline{x_{\tau_3,a_1,a_2,j_1}} \lambda_{\tau_3} S_2^\dagger}{\sqrt{3}}$		
								$-h \log \left(\frac{\mu b m_2^2}{M_F^2} \right) \overline{y_{1,i_1,\tau_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{2,i_1,\tau_1} \overline{x_{\tau_3,a_1,a_2,j_1}} \lambda_{\tau_3} S_2^\dagger$		
								$-2h \log \left(\frac{\mu b m_2^2}{M_F^2} \right) \overline{y_{2,i_1,\tau_1}} \overline{y_{e,i_2,i_2}} H_{a_1} S_2 y_{2,i_2,\tau_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger$		
								$-\frac{h \log \left(\frac{\mu b m_2^2}{M_F^2} \right) \overline{y_{1,i_1,\tau_1}} \overline{\lambda_{\tau_3}} S_2 y_{1,i_1,\tau_1} \text{Bn r(T3 Mix CG)}_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$		
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-h \log \left(\frac{\mu b m_2^2}{M_F^2} \right) \overline{y_{2,i_1,\tau_1}} \overline{\lambda_{\tau_3}} S_2 y_{2,i_1,\tau_1} \text{Bn r(T3 Mix CG)}_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$		
								$-2h \log \left(\frac{\mu b m_2^2}{M_F^2} \right) \overline{y_{2,i_1,\tau_1}} H_{a_2} S_2 y_{2,i_1,\tau_1} H_{a_1}^\dagger S_2^\dagger \overline{x_{\text{H2},a_1,a_2}^1} \lambda_{\text{H2}^1}$		